

What Is My Child Actually Learning?

A Parent's Guide to Data, Coding & AI

FREE EXCERPT

What Is AI? (And What It Isn't)

Module 4 — including the honest guide to AI and homework that most schools aren't giving parents.

ABOUT THIS EXCERPT

This is a free excerpt from *What Is My Child Actually Learning?* — a plain-English guide for parents who want to understand what their children are being taught in coding, data, and AI. No technical background required.

Module 4 covers the topic parents ask about most: *artificial intelligence*. What it actually is, how it works, what your child is learning about it — and the question that comes up at almost every parents' evening right now: what should I do about AI and homework?

What Is AI? (And What It Isn't)

If you've used ChatGPT, seen a deepfake, or watched a recommendation algorithm send you down a YouTube rabbit hole, you've already experienced AI. But most parents — and many teachers — don't have a clear picture of what's actually happening underneath. This module gives you that picture. Not the science-fiction version, not the breathless tech-press version — just a clear, honest explanation of what AI is, what it can and can't do, and what your child is actually learning about it at school.

The one-sentence explanation

AI systems find patterns in large amounts of data, and use those patterns to make predictions. That's it. That's the core of it. Everything else — ChatGPT, image generators, facial recognition, fraud detection — is a variation on that same idea.

When ChatGPT writes a sentence, it's predicting which word is most likely to come next, based on patterns it found in billions of pieces of text. When Spotify recommends a song, it's predicting what you'll like based on patterns in what millions of other listeners played. When your bank flags a suspicious transaction, it's predicting whether a payment matches the pattern of fraud.

THE KEY WORD IS "PREDICTING"
AI doesn't understand, reason, or think. It predicts. That distinction matters — because it explains both what AI is remarkably good at and where it goes badly wrong.

What AI is good at — and where it fails

Understanding AI's strengths and weaknesses will help you have a much more grounded conversation with your child than most adults can manage.

What AI does well	Where AI fails or misleads
Recognising patterns in huge datasets	Understanding context the way humans do
Generating fluent, confident-sounding text	Knowing whether that text is actually true
Translating between languages	Grasping nuance, irony, or cultural subtext
Spotting anomalies (fraud, medical scans)	Explaining its own reasoning clearly

What AI does well	Where AI fails or misleads
Recommendations (music, films, products)	Dealing with situations outside its training data

The most important thing in the right-hand column: AI generates confident-sounding text that is sometimes completely wrong. This is called "hallucination" and it's not a bug that will be fixed — it's a fundamental characteristic of how these systems work.

AI is very good at sounding right. It is not always good at being right. Your child needs to know the difference.

What your child is actually learning about AI

This varies significantly by age and school, but here's a broad picture:

PRIMARY SCHOOL (AGES 7–11)

At this age, children learn that computers follow instructions (algorithms), and begin to understand that technology makes decisions based on rules. They're unlikely to use the word "AI" formally, but they're building the logical thinking that underpins it.

SECONDARY SCHOOL (AGES 11–16)

Students learn what machine learning is and how it differs from traditional programming. They explore bias in AI systems — why a facial recognition system might perform worse on darker skin tones, for example — and discuss the ethical implications. Some will build simple AI models themselves.

The emphasis on *AI ethics and bias* is deliberate and important. Schools aren't just teaching children to use AI tools — they're teaching them to question those tools. That's a more valuable skill than knowing which button to press.

THE QUESTION EVERY PARENT IS ASKING

AI and homework: the honest answer

Let's be direct. Your child almost certainly has access to ChatGPT or similar tools, whether you've given them permission or not. The question isn't whether they're using AI — it's whether they're using it well.

Schools are struggling with this too. Policies range from blanket bans (largely unenforceable) to enthusiastic adoption (sometimes uncritical). Most parents are somewhere in the middle, unsure what to think. Here is a framework that's more useful than either extreme:

THE PROBLEM WITH BANNING IT

AI tools are already part of professional life. A blanket ban at home doesn't teach your child to use them responsibly — it just means they use them without your guidance. It's a bit like banning calculators: the skill isn't avoiding the tool, it's knowing when and how to use it.

THE PROBLEM WITH ALLOWING EVERYTHING

If your child submits AI-written work as their own, two things happen: they don't develop the underlying skill the homework was meant to build, and they risk serious academic consequences if caught. Most schools now use AI detection tools, and the penalties are significant.

A more useful way to think about it

The most helpful question isn't "did they use AI?" but *what did they learn?* Consider the difference between these two scenarios:

SCENARIO A – USING AI AS A SHORTCUT

Your child has an essay due. They type the question into ChatGPT, copy the output, and submit it. They haven't thought about the topic, haven't practised writing, and haven't developed their own argument. The essay might be decent. The learning was zero.

SCENARIO B – USING AI AS A THINKING PARTNER

Your child drafts their own argument first. They ask ChatGPT to challenge their reasoning or suggest counterarguments. They use the response to strengthen their own thinking, then write the final essay themselves. The submission is their own work — AI helped them think more rigorously.

Scenario B is legitimate. Scenario A is not — and most schools draw the line there. The distinction is: *did the AI do the thinking, or did it support the thinking?*

How to talk to your child about this

The goal isn't to interrogate — it's to open a conversation. Here are five questions that tend to work:

- 1 "Have you tried using ChatGPT or any AI tool for schoolwork?" Opens the conversation without accusation. Most children will be honest if they don't feel cornered.
- 2 "What do you think it's actually good at? Where does it get things wrong?" Shifts into curiosity rather than judgement. Gets them thinking critically.
- 3 "Does your school have a policy on using AI for homework?" Puts the rules in the school's mouth, not yours. Useful if they're fuzzy on what's allowed.
- 4 "If you used AI to help with something, what did you actually learn from it?" The key question. It's not about whether they used it — it's about what they got from it.
- 5 "What would you do if AI gave you a confident answer that turned out to be wrong?" Tests whether they're thinking critically about AI outputs. The answer is revealing.

The deeper point worth making to your child

AI tools are going to be part of professional life. The adults who thrive won't be those who avoided AI — they'll be those who learned to use it critically, direct it purposefully, and check its outputs carefully.

But here's the thing: you can't direct an AI tool well on a subject you don't understand. A student who uses AI to avoid doing the thinking is skipping exactly the part that makes AI useful later. The underlying knowledge still matters — possibly more than ever.

THE ANALOGY WORTH SHARING WITH YOUR CHILD

A calculator is extraordinarily useful — if you understand maths well enough to know when the answer looks wrong. AI is the same. The tool is powerful. The judgement that makes it useful still has to come from you.

TRY THIS WEEK

Module 4 Activity: Five questions, one conversation

This week, pick one of the five questions above and ask your child. Not as a test — as genuine curiosity. Listen to the answer before responding. You might be surprised what they already know.

THE COMPLETE GUIDE

Want the full guide?

This is a free excerpt from *What Is My Child Actually Learning?* — a complete parent's guide to coding, data, and AI in schools.

Seven plain-English modules · Read on any device · No technical knowledge needed

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The honest case for why coding and data matter — with real numbers on jobs and salaries.

02 What Is Coding, Really?

Demystifying programming. What Scratch and Python actually are, and the myths addressed directly.

03 What Is Data and Why Does It Matter?

How apps, recommendations, and social media work. Your child's digital footprint.

04 What Is AI? — you are here

This module. The honest guide to AI and homework.

05 Digital Skills: Staying Safe and Thinking Critically

Online safety, misinformation, and lateral reading.

06 How to Support Your Child

How to ask the right questions and spot if they're struggling — without knowing the answers yourself.

07 Try It Yourself

A 30-minute Scratch session for parents. Feel what your child feels when learning something new.